

1 Competition: Autarkies (7(+7) Points)

Write an anytime algorithm that computes a maximal autarky for a given formula, i.e., an autarky that satisfies as many clauses as possible. The tool should accept a DIMACS file path as input and, as soon as possible, output a set of literals representing an autarky. Then, it should output improved sets iteratively. For each autarky, use a single line that begins with "v " and ends with a zero, with space-separated literals of the autarky in between. This format is similar to the SAT Competition Solution Output Format [1].

We will run your submission on DIMACS files and evaluate the quality of the solutions found within a set time limit. We will stop each run after k minutes. Then, we will check the last output line to see if it contains a valid autarky and compute the number of clauses satisfied by the autarky. The solution that finds the best autarkies on average will win the competition.

[1] **Output Format:** <https://satcompetition.github.io/2025/output.html>

2 Variable Elimination (3 Points)

For a gate encoding E with output x in a formula $F = E \cup R$, we simplified the resolvents $(E_x \cup R_x) \otimes (E_{\bar{x}} \cup R_{\bar{x}})$ by $S := (E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}})$, dropping both $R_x \otimes R_{\bar{x}}$ and $E_x \otimes E_{\bar{x}}$. Show that the clauses in $R_x \otimes R_{\bar{x}}$ can be derived from S by resolution. You can assume that E encodes a binary AND gate.

3 Variable Elimination (2×2 Points)

Let the formula S with gate encodings E_1 and E_2 be given. Apply variable elimination for gates for variables a and r . Give the clause sets after each elimination step. Try the following two strategies.

1. Eliminate variable a first, and then r if possible.
2. Eliminate variable r first, and then a if possible.

$$S = \underbrace{\{\{\neg x, \neg y, a\}, \{x, \neg a\}, \{y, \neg a\}\}}_{E_1} \underbrace{\{\{\neg a, r\}, \{\neg z, r\}, \{a, z, \neg r\}, \{a, z, r\}, \{\neg a, \neg r\}\}}_{E_2}$$

4 Blocked Clauses (3×3 Points)

If Blocked Clause Elimination (BCE) reduces a formula F to the empty formula then F is called a blocked set. Prove the following statements.

1. Any formula F can be partitioned into two blocked sets S and L such that $F = S \cup L$. Design a linear algorithm that produces L and S from F .
2. Blocked sets are not closed under resolution. If F is a blocked set then $F \cup C_1 \otimes C_2$, where $C_1, C_2 \in F$ may not be a blocked set anymore.
3. Blocked sets are not closed under partially assigning variables. If F is a blocked set then $F_{x=v}$ (the result of assigning v to x and subsequent simplification) may not be a blocked set anymore.

5 Competition: Clause Sharing Selection Heuristic (3(+7) Points)

Manipulate Mallob's clause sharing by modifying the class `ClausePrefilter` (see below), whose method `prefilterClause` is called whenever a solver thread produces a clause that passes a number of basic checks. The method can return `false` to completely drop a clause; if the method returns `true`, the calling solver thread will subsequently attempt to write the clause to an export buffer. You can change a (non unit) clause's priority by manipulating its LBD value to make it more or less likely that a clause "survives" subsequent selection stages and is shared successfully. For a clause of length $\ell \geq 2$, LBD values of 2 ("best") through ℓ ("worst") are valid.

If needed, you may perform precomputations and access the formula in the body of `notifyFormula()`.

You get 3 points for submitting a functional version of `clause_prefilter.hpp` with some non-trivial change. The submissions that result in the best performance with a Kissat portfolio at 64 cores on diverse benchmarks get up to 7/5/3 additional points. We will, in particular, set the non-default parameters `-lbdpi=1 -lbdpo=1 -pbbm=1` to consistently prioritize clauses by “LBD” first and by clause length second (i.e., as a tiebreaker only) and `-rlbd=1 -ilbd=0` to reset each incoming clause’s LBD value to ℓ before a solver thread imports it.

Mallob version to use: <https://github.com/domschrei/mallob/tree/experimental>

— `ClausePrefilter`: see `src/app/sat/sharing/filter/clause_prefilter.hpp`

Contact Dominik (dominik.schreiber@kit.edu) in case of technical questions or issues.

6 Parallel Resolution (3 + 3 Points)

Consider a parallel setup with p resolution procedures in a “bulk synchronous parallel” (BSP) model: Each procedure performs a single resolution step; then, all resolvents are shared across all procedures. This is repeated until the empty clause is found.

- (a) Provide an unsatisfiable CNF formula where the above setup *can* speed up the derivation of the empty clause by a factor of $p/\log p$ (or better) compared to the shortest sequential resolution order.
- (b) Provide an unsatisfiable CNF formula where the derivation of the empty clause can only ever be sped up by a factor of $1 + \varepsilon$ (where $0 \leq \varepsilon \ll 1$) compared to the shortest sequential resolution order.

7 VBS and Speedups (4 + 6 Points)

Analyze the results of the 2024 SAT Competition.

- (a) Compute a VBS of all sequential main track solvers. Compare its performance to the best cloud track solver in the competition. Interpret the results, also considering any peculiar instance families.
- (b) Compute the geometric mean and median speedup of each parallel track solver relative to the best performing sequential solver `kissat-sc2024`, only considering instances which both the sequential solver *and* the parallel solver were able to solve. Then do the same while considering *all* instances solved by the parallel solver – “generously” attributing a running time of 5 000 s to each such instance that the sequential solver did not solve. Interpret and compare the results.

Competition results: https://satcompetition.github.io/2024/downloads/detailed_results.zip

Benchmark meta data: <https://satcompetition.github.io/2024/downloads/meta.csv>

8 Clause Filtering in Distributed SAT (5 Points)

Distributed clause-sharing solvers like MALLOBSAT filter a repeated clause based on *exact syntactical equivalence* (i.e., if a clause c is shared successfully, then clause c will be blocked for some period). Find a way to generalize this approach to *subsumed* clauses, i.e., if clause c is shared, then all clauses $c' \supseteq c$ are blocked for some period. Provide a rough analysis of the running time complexity and the memory footprint of your approach.